Matteo Sesia

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Citizenship: USA, Italy

Research Areas

Statistical methods for uncertainty quantification, variable selection, and reproducible data analysis; distribution-free and model-agnostic inference; machine learning and applied statistics.

Academic Positions

University of Southern California, Marshall School of Business:

Associate Professor of Data Sciences and Operations, tenured (04/2025–present)
Kenneth King Stonier Assistant Professor of Business Administration (05/2024–04/2025)
Assistant Professor of Data Sciences and Operations, tenure-track (06/2020–04/2025)

University of Southern California, Viterbi School of Engineering:

Associate Professor of Computer Science, by courtesy (04/2025–present)
Assistant Professor of Computer Science, by courtesy (01/2023–04/2025)

Education

Ph.D. in Statistics, Stanford University (2020). Advisor: Emmanuel Candès.
Thesis title: New methods for variable importance testing with applications to genetic studies.*

M.S. in Physics of Complex Systems, Politecnico di Torino and Université Paris-Sud (2015).
Graduated *cum laude* (highest honors).

M.A. in Statistics and Applied Mathematics, Collegio Carlo Alberto (2015).
Graduated *with distinction* (highest honors).

B.S. in Engineering Physics, Politecnico di Torino (2013).
Graduated *cum laude* (highest honors).

Editorial Appointments

Associate Editor, *Journal of the Royal Statistical Society – Series B (Methodological)* (11/2025–present).

Associate Editor, *Biometrika* (08/2025–present).

Funding and awards

Alexey Chervonenkis Best Paper Award, COPA 2025.

Alexey Chervonenkis Best Poster Award, COPA 2025.

Google Research Scholar Award, 2025. ($60,000)

USC-Capital One Center for Responsible AI Decision Making in Finance, (PI), 2025. ($84,000)

Top Reviewer, NeurIPS 2024

GenAI Research Grant (co-PI), University of Southern California, 2023–2024. ($100,000)

Dr. Douglas Basil Award for Junior Business Faculty, USC Marshall School of Business, 2023.

National Science Foundation Grant DMS 2210637 (PI), 2022–2025. ($160,000)

Amazon Research Award, 2022. ($70,000 + $50,000 in computing credits)

*Jerome H. Friedman Applied Statistics Dissertation Award, Stanford University, 2020.

Consulting and Expert Work Activities

2024–present — Providing long-term consulting for a major pharmaceutical company on uncertainty quantification and reproducibility in predictive modeling for biomedical and survival analysis applications.

2024 — Served on a three-member academic panel to evaluate the merit and feasibility of a proposed Ph.D. program in Data Science, advising a state-level higher education authority.

2022–2023 — Advised a book author on statistical methodology, including cleaning and analysis of nationwide crime victimization survey data and official crime reports, leading to a published work.

2021–2022 — Consulted with academic collaborators on multiple projects leading to peer-reviewed publications in biomedicine and healthcare journals.

2017–2019 — Participated in, and for one term led, the Stanford Statistics Consulting Workshop, providing pro bono statistical support to researchers across the university as part of a team of Ph.D. students.

Other Professional Experiences

Visiting scholar, Collegio Carlo Alberto (Torino, Italy). (05/2023)

Research intern, Adobe Inc. (San Jose, California). (06/2017–08/2017)

Publications and Preprints

Publications in refereed journals (statistics and general-interest)

[1] Z. Liang, T. Xie, X. Tong, M. Sesia. Structured conformal inference for matrix completion with applications to group recommender systems. *Journal of the American Statistical Association* (2026) https:/arxiv.org/abs/2404.17561

[2] P. Gablenz, M. Sesia, T. Sun, C. Sabatti. Searching for local associations while controlling the false discovery rate. *Journal of the American Statistical Association* (2026) https://doi.org/10.1080/01621459.2026.2616856

[3] M. Beraha, S. Favaro, M. Sesia. A smoothed-Bayesian approach to frequency recovery from sketched data. *Journal of the American Statistical Association* (2025) https://doi.org/10.1080/01621459.2025.2490302

[4] M. Sesia, Y. X. Wang, X. Tong. Adaptive conformal classification with noisy labels. *Journal of the Royal Statistics Society Series B (Statistical Methodology)* (2024) https://doi.org/10.1093/jrsssb/qkae114

[5] M. Beraha, S. Favaro, M. Sesia. Random measure priors in Bayesian recovery from sketches. *Journal of Machine Learning Research* (2024) https://jmlr.org/papers/v25/23-1058.html

[6] Z. Liang, M. Sesia, W. Sun. Integrative conformal p-values for powerful out-of-distribution testing with labeled outliers. *Journal of the Royal Statistics Society Series B (Statistical Methodology)* (2024) https://doi.org/10.1093/jrsssb/qkad138

[7] M. Sesia, S. Favaro, E. Dobriban. Conformal frequency estimation using discrete sketched data with coverage for distinct queries. *Journal of Machine Learning Research* (2023) https://jmlr.org/papers/v24/22-1278.html

[8] S. Bates, E. Candès, L. Lei, Y. Romano, M. Sesia. Testing for outliers with conformal p-values. *Annals of Statistics* (2023). https://doi.org/10.1214/22-aos2244

[9] S. Li, Z. Ren, C. Sabatti, M. Sesia. Transfer learning in genome-wide association studies with knockoffs. *Sankhya B* (2022) https://doi.org/10.1007/s13571-022-00297-y

[10] S. Li, M. Sesia, Y. Romano, E. Candès, C. Sabatti. Searching for robust associations with a multi-environment knockoff filter. *Biometrika* (2021). https://doi.org/10.1093/biomet/asab055

[11] M. Sesia, S. Bates, E. Candès, J. Marchini, C. Sabatti. False discovery rate control in genome-wide association studies with population structure. *Proceedings of the National Academy of Sciences* (2021). https://doi.org/10.1073/pnas.2105841118

[12] S. Bates, M. Sesia, C. Sabatti, E. Candès. Causal inference in genetic trio studies. *Proceedings of the National Academy of Sciences* (2020). https://doi.org/10.1073/pnas.2007743117

[13] M. Sesia, E. Katsevich, S. Bates, E. Candès, C. Sabatti. Multi-resolution localization of causal variants across the genome. *Nature Communications* (2020). https://doi.org/10.1038/s41467-020-14791-2

[14] M. Sesia, E. Candès. A comparison of some conformal quantile regression methods. *Stat* (2020). http://dx.doi.org/10.1002/sta4.261

[15] Y. Romano, M. Sesia, E. Candès. Deep knockoffs. *Journal of the American Statistical Association* (2019). https://doi.org/10.1080/01621459.2019.1660174

[16] M. Sesia, C. Sabatti, E. Candès. Gene hunting with hidden Markov model knockoffs. *Biometrika* (2019). https://doi.org/10.1093/biomet/asy033.
Discussion paper with rejoinder https://doi.org/10.1093/biomet/asy075.

Publications in refereed conferences (computer science)

[17] M. Sesia, V. Svetnik. Conformal survival bands for risk screening under right-censoring. *14th Symposium on Conformal and Probabilistic Prediction with Applications* (2025)
https://proceedings.mlr.press/v266/sesia25a.html (Best Paper Award)

[18] M. Sesia, V. Svetnik. Doubly robust conformalized survival analysis with right-censored data. *ICML* (2025) https://icml.cc/virtual/2025/poster/46585 (selected as "Spotlight", $< 3\%$ acceptance rate)

[19] M. Bashari, M. Sesia, Y. Romano. Robust conformal outlier detection under contaminated reference data. *ICML* (2025) https://icml.cc/virtual/2025/poster/43852

[20] Y. Zhou, M. Sesia. Conformal classification with equalized coverage for adaptively selected groups *NeurIPS* (2024) https://arxiv.org/abs/2405.15106

[21] X. Huang, S. Li, M. Yu, M. Sesia, H. Hassani, I. Lee, O. Bastani, E. Dobriban. Uncertainty in language models: assessment through rank-calibration. *Conference on Empirical Methods in Natural Language Processing* (2024) https://doi.org/10.18653/v1/2024.emnlp-main.18

[22] Y. Zhou, L. Lindemann, M. Sesia. Conformalized adaptive forecasting of heterogeneous trajectories. *ICML* (2024) https://proceedings.mlr.press/v235/zhou24l.html

[23] M. Bashari, A. Epstein, Y. Romano, M. Sesia. Derandomized novelty detection with FDR control via conformal e-values. *NeurIPS* (2023) https://arxiv.org/abs/2302.07294

[24] Z. Liang, Y. Zhou, M. Sesia. Conformal inference is (almost) free for neural networks trained with early stopping. *ICML* (2023) https://proceedings.mlr.press/v202/liang23i.html

[25] B. Einbinder, Y. Romano, M. Sesia, Y. Zhou. Training uncertainty-aware classifiers with conformalized deep learning. *NeurIPS* (2022) https://arxiv.org/abs/2205.05878

[26] M. Sesia, S. Favaro. Conformal frequency estimation with sketched data. *NeurIPS* (2022)
https://arxiv.org/abs/2204.04270

[27] N. Fingerhut, M. Sesia, Y. Romano. Coordinated double machine learning. *ICML* (2022)
https://proceedings.mlr.press/v162/fingerhut22a.html

[28] M. Sesia, Y. Romano. Conformal prediction using conditional histograms. *NeurIPS* (2021)
https://arxiv.org/abs/2105.08747 (selected as "Spotlight", $< 3\%$ acceptance rate)

[29] Y. Romano, M. Sesia, E. Candès. Classification with valid and adaptive coverage. *NeurIPS* (2020) https://arxiv.org/abs/2006.02544 (selected as "Spotlight", $< 3\%$ acceptance rate)

Publications in refereed journals (biomedicine and healthcare)

[30] M. A. Juratli, D. Roy, E. Oppermann, M. Sesia, A. Schnitzbauer, J. Hoelzen, S. Katou, H. Morgul, B. Strucker, A. Pascher, M. Heikenwalder, W. O. Bechstein. Long-term effect of liver resection on circulating stem cells expressing PD-L1 in patients with hepatocellular carcinoma – Pilot study. *Zeitschrift für Gastroenterologie* (2022) https://www.thieme-connect.com/products/ejournals/abstract/10.1055/s-0042-1754779

[31] M. A. Juratli, D. Roy, E. Oppermann, M. Sesia, A. Schnitzbauer, J. Hoelzen, S. Katou, H. Morgul, B. Strucker, A. Pascher, W. O. Bechstein. Long-term dynamics of circulating tumor cells and their prognostic relevance after liver resection in patients with hepatocellular carcinoma. *Zeitschrift für Gastroenterologie* (2022) https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0042-1754884

[32] J. Hoelzen, K. Sander, M. Sesia, D. Roy, E. Rijcken, A. Schnabel, B. Strücker, M. Juratli, A. Pascher. Robotic-assisted esophagectomy leads to significant reduction in postoperative acute pain: A retrospective clinical trial. *Annals of Surgical Oncology* (2022) https://doi.org/10.1245/s10434-022-12200-0.

[33] A. Fayazi, M. Sesia, K. S. Anand. Hyperoxemia among pediatric intensive care unit patients receiving oxygen therapy. *Journal of Pediatric Intensive Care* (2021). https://doi.org/10.1055/s-0041-1740586

[34] C. Chia, M. Sesia, C.-S. Ho, S. Jeffrey, J. Dionne, E. Candès, R. Howe. Interpretable classification of bacterial Raman spectra with knockoff wavelets. *IEEE Journal of Biomedical and Health Informatics* (2021). https://doi.org/10.1109/JBHI.2021.3094873

Pre-print manuscripts (under review or revision)

[35] M. Sesia, S. Favaro. Elements of Conformal Prediction for Statisticians. (2026) Invited submission to *Annual Review of Statistics and Its Application.* https://arxiv.org/abs/2603.23923

[36] M. Sesia, V. Svetnik. Distribution-free selection of low-risk oncology patients for survival beyond a time horizon. (2025) https://arxiv.org/abs/2512.18118

[37] Y. Fan, M. Sesia. Interpretable multivariate conformal prediction with fast transductive standardization. (2025) https://arxiv.org/abs/2512.15383

[38] T. Xie, Y. Zhou, Z. Liang, S. Favaro, M. Sesia. Conformal inference for open-set and imbalanced classification. (2025) https://arxiv.org/abs/2510.13037

[39] Y. Zhao, X. Yu, M. Sesia, J. Deshmukh, L. Lindemann. Conformal predictive programming for chance constrained optimization. (2025) https://arxiv.org/abs/2402.07407

[40] T. Bortolotti, Y. X. R. Wang, X. Tong, A. Menafoglio, S. Vantini, M. Sesia. Noise-adaptive conformal classification with marginal coverage. (2025) https://arxiv.org/abs/2501.18060

[41] C. Magnani, M. Sesia, A. Solari. Collective outlier detection and enumeration with conformalized closed testing. (2024) https://arxiv.org/abs/2308.05534

Patents

M. Sesia and Y. Abbasi-Yadkori (Adobe Inc). "Recommendation system using linear stochastic bandits and confidence interval generation". US 11,100,559. August 24, 2021.

Teaching

University of Southern California:

Data Science for Business (GSBA 524), graduate. Spring 2026.

Data Science for Business (GSBA 524), graduate. Spring 2025.

Data Science for Business (GSBA 524), graduate. Spring 2024.
Applied Time Series Analysis for Forecasting (DSO 522), graduate. Fall 2024.
Applied Managerial Statistics (GSBA 506b), graduate. Spring 2023.
Applied Business Statistics (BUAD 310), undergraduate. Fall 2021.
Applied Business Statistics (BUAD 310), undergraduate. Fall 2020.

Università di Torino and Collegio Carlo Alberto:
Predictive uncertainty in ML with conformal inference, graduate. May 2023.

Stanford University:
Consulting Workshop (Stats 390), graduate. Summer 2018.
Introduction to R (Stats 195), undergraduate. Spring 2018, Spring 2020.

Professional Service

Journal referee: Annals of Applied Statistics, Annals of Statistics, Bayesian Analysis, Biometrika, Briefings in Bioinformatics, Electronic Journal of Statistics, Human Genetics, Information Sciences, Journal of Computational and Graphical Statistics, Journal of Machine Learning Research, Journal of the American Statistical Association, Journal of the Royal Statistical Society, Machine Learning, Nature Communications, Operations Research, SIAM Journal on Mathematics of Data Science, Statistics and Computing, Statistics and Probability Letters, Statistics in Medicine, Statistical Science.

Conference referee: COLT, COPA, ISIT, NeurIPS, ICML.

Ad-hoc grant reviewer: Israel Science Foundation 2022, 2024. Israel Ministry of Innovation, Science and Technology, 2024. Italian Ministry of Research, 2026.

Grant review panelist: National Science Foundation 2023, 2024.

Conference session organizer: SEEDS (2024), WNAR (2024).

Internal Service

Statistics seminar organizer (2021–2022, 2022–2023); faculty hiring committee (2021–2022, 2022–2023); PhD admission committee (2020–2021, 2021–2022).

Executive committee for the Inaugural 2024 SEEDS conference.

Dissertation committee: Gregory Faletto (Data Sciences and Operations), Jacob Gizamba (Population, Health and Place). Qualifying exam committee: Yiqiu Shen (Data Sciences and Operations).

Students

Current students: Tianmin Xie (PhD, Data Sciences and Operations), Yanfei Zhou (PhD, Data Sciences and Operations), Brian Fan (PhD, Mathematics).

Past students: Ziyi Liang (PhD in Mathematics, 2024). Now post-doctoral researcher at UC Irvine.

Presentations

Keynote lectures
Conformal and Probabilistic Prediction with Applications. 09/09/2024, in Milan, Italy.

Invited seminar presentations
University of California - Los Angeles, Department of Statistics. 12/04/2025.
University of Chicago, Department of Statistics. 12/01/2025.
Harvard University, Department of Statistics. 04/22/2024.
University of Pennsylvania, Department of Statistics and Data Science. 04/17/2024.
University of Washington, Department of Biostatistics. 11/09/2023.

University of Pittsburgh, Department of Statistics. 10/23/2023.
Polytechnic University of Milan, Department of Mathematics. 09/21/2023.
Università Cattolica, Department of Statistical Sciences. 05/18/2023.
Collegio Carlo Alberto, Statistics Seminar. 05/11/2023,
University of California - Irvine, Department of Mathematics. 03/06/2023.
University of Southern California, Department of Computer Science, 10/27/2022.
University of California - Santa Cruz, Department of Statistics. 10/17/2022.
University of Southern California, Department of Economics, 02/25/2022.
Yale University, School of Public Health. 02/16/2021.
University of Nottingham, School of Mathematical Sciences. 12/17/2020.
University of Milan - Bicocca, DEMS Department. 12/02/2020.
Johns Hopkins University, Mathematical Institute for Data Science. 02/18/2020.
University of Southern California, DSO Department. 01/27/2020.
University of California - Davis, Department of Statistics. 01/06/2020.
Stanford University, Statistics Department. 07/16/2019.
Collegio Carlo Alberto, Statistics Seminar. 12/19/2018.
Université Grenoble Alpes, Bayes in Grenoble Seminar. 07/10/2018.

Invited conference and workshop presentations

Open SkyAI Conference. 09/04/2025, in Chicago, IL.
ICML Expo Workshop on AI in Finance. 07/14/2025, in Vancouver, Canada.
Computational Genomics Summer Institute, 07/10/2025, in Palos Verdes, CA.
SIAM MDS 2024, mini-symposium on Mathematics of Responsible ML, 10/21/2024, in Atlanta, GA.
WNAR of the International Biometric Society, 06/12/2024, in Fort Collins, CO.
International Conference on Statistics and Data Science, 12/18/2023, in Lisbon, Portugal.
Joint Statistical Meeting, Invited Poster Session. 08/06/2023, in Toronto, Canada.
Computational Genomics Summer Institute, 07/15/2023, in Big Bear Lake, CA.
Boston University, New England Statistics Symposium (short course). 06/04/2023, in Boston, MA.
National Institute of Statistical Sciences. 04/19/2023, in Washington, D.C. (remote).
International Conference on Statistics and Data Science, 12/13/2022, in Florence, Italy.
SIAM Symposium on Mathematics of Interpretable ML, 09/30/2022, in San Diego, CA.
Computational Genomics Summer Institute, 07/07/2022, in Big Bear Lake, CA.
International Seminar on Selective Inference, 06/02/2022, (remote).
Stanford University, Statistics Industrial Affiliates Meeting. 02/22/2019, in Stanford, CA.

Invited industry presentations

Merck & Co., Inc. 02/10/2022, in Morristown, NJ (remote).
Regeneron Pharmaceuticals, Inc. 09/17/2019, in Eastview, NY.
23andMe, Inc. 05/21/2019, in Mountain View, CA.

Invited discussions

International Seminar on Selective Inference, Nov. 15/2023, (remote).
International Seminar on Selective Inference, Nov. 17/2022, (remote).
International Seminar on Selective Inference, 10/22/2020, (remote).

Contributed conference and workshop presentations

Joint Statistical Meeting, Topic-Contributed Paper Session. 08/09/2023, in Toronto, Canada.
RMDS Lab, IM DATA Conference 2022, 08/14/2022, in Los Angeles, CA.
ICML, spotlight presentation. 07/19/2022, in Baltimore, MD.

NeurIPS, spotlight presentation. 12/09/2021 (remote).
RMDS Lab, IM DATA Conference, 10/28/2021, in Pasadena, CA (remote).
RMDS Lab, IM DATA Conference, 11/02/2020, in Pasadena, CA (remote).
NeurIPS, spotlight presentation. 12/08/2020 (remote).
Royal Statistics Society Conference, 09/04/2018, in Cardiff, United Kingdom.
Workshop on Model Selection, Regularization and Inference, 07/13/2018, in Vienna, Austria.
Computational and Methodological Statistics Conference, 12/17/2017, in London, UK.

Contributed poster presentations
ICML, 07/15/2025-07/16/2025, in Vancouver, Canada.
NeurIPS, 12/13/2024, in Vancouver, Canada.
ICML, 07/19/2022, in Baltimore, MD.
American Society for Human Genetics Annual Meeting, 10/20/2021, virtual meeting.
ICML Workshop on distribution-free uncertainty quantification, 07/24/2021, virtual meeting.
American Society for Human Genetics Annual Meeting, 10/28/2020, virtual meeting.
American Society for Human Genetics Annual Meeting, 10/16/2019, in Houston, TX.
Higher-Order Asymptotics and Post-Selection Inference Workshop, 08/18/2019, in St. Louis, MO.
American Society for Human Genetics Annual Meeting, 10/18/2018, in San Diego, CA.